CSA0201 – C PROGRAMMING

Smart Emergency Resource Management and Analysis System using C

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IMPLEMENTATION RESULTS

The screenshots below were captured from the Google Colab notebook used to build and run resource_management.c. The first cells write the C source file to disk in parts (%%writefile / %%writefile -a); the final cell compiles it with GCC and runs the compiled program (!gcc -Wall -o resmgmt resource_management.c && ./resmgmt), exercising all 13 menu options against a 5-resource sample data set spanning the Emergency, ICU, and Trauma departments.

Screenshot 1: *Cell 1 — writing the structure definition, prototypes, and main() menu driver*

```
[1]
%%writefile resource_management.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 100
#define FILENAME "resources.txt"
#define REPORTFILE "report.txt"

typedef struct {
    int  id;
    char name[50];
    char category[30];
    char department[30];
    int  quantity;
    int  threshold;
    int  priority;
} Resource;

Resource resources[MAX];
int resCount = 0;

void  printMenu();
void  addResource();
void  updateResource();
void  displayAll(Resource arr[], int n);

void  searchMenu();
int   searchByID(Resource arr[], int n, int id);
int   searchByName(Resource arr[], int n, char *name);
int   searchByCategory(Resource arr[], int n, char *category, int result[]);
int   recursiveBinarySearchByID(Resource arr[], int low, int high, int id);

void  sortMenu();
void  swap(Resource *a, Resource *b);
void  bubbleSortByQuantity(Resource *arr, int n);
void  selectionSortByPriority(Resource *arr, int n);
void  sortByDepartment(Resource *arr, int n);

void  mergeDepartments();
```

```
Overwriting resource_management.c
```

Screenshot 2: *Cell 1 (continued) — main() switch-case dispatch and printMenu()*

```
[1]
int   isDuplicate(Resource arr[], int n, Resource r);

void  findDuplicates();
void  findDuplicatesRecursive(int index, int *found);

void  analyseResources();
void  displayCriticalResources();

double totalQuantityRecursive(Resource arr[], int n);
void   departmentWiseTotalRecursive(char depts[][30], int deptCount, int index);
void   generateReport();

void  saveToFile();
void  loadFromFile();

int main() {
    loadFromFile();
    int choice;

    do {
        printMenu();
        printf("Enter your choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input! Please enter a number.\n");
            while (getchar() != '\n');
            continue;
        }

        switch (choice) {
            case 1:  addResource();                 break;
            case 2:  updateResource();               break;
            case 3:  displayAll(resources, resCount); break;
            case 4:  searchMenu();                   break;
            case 5:  sortMenu();                     break;
            case 6:  mergeDepartments();              break;
            case 7:  findDuplicates();                break;
            case 8:  analyseResources();              break;
            case 9:  displayCriticalResources();      break;
            case 10: generateReport();                break;
            case 11: saveToFile();                    break;
            case 12: loadFromFile();                  break;
            case 13: printf("Saving data and exiting...\n"); saveToFile(); break;
            default: printf("Invalid choice! Please try again.\n");
        }
        printf("\n");
    } while (choice != 13);

    return 0;
}

void printMenu() {
    printf("\n===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====\n");
    printf(" 1. Add New Resource\n");
    printf(" 2. Update Resource Details\n");
    printf(" 3. Display All Resources\n");
    printf(" 4. Search Resource (ID / Name / Category)\n");
    printf(" 5. Sort Resources (Quantity / Priority / Department)\n");
    printf(" 6. Merge Resources from Two Departments\n");
    printf(" 7. Identify Duplicate Resources\n");
    printf(" 8. Analyse Resource Availability\n");
    printf(" 9. Display Critical / Low-Stock Resources\n");
    printf("10. Generate Consolidated Report\n");
    printf("11. Save Records to File\n");
    printf("12. Retrieve Records from File\n");
    printf("13. Exit\n");
    printf("======================================================\n");
}
```

```
Overwriting resource_management.c
```

Screenshot 3: *Cell 2 — addResource() and updateResource() functions*

```
[2]
%%writefile -a resource_management.c

void addResource() {
    if (resCount >= MAX) {
        printf("Resource storage is full!\n");
        return;
    }

    Resource r;
    printf("Enter Resource ID: ");
    scanf("%d", &r.id);

    if (searchByID(resources, resCount, r.id) != -1) {
        printf("Error: a resource with this ID already exists!\n");
        return;
    }

    printf("Enter Resource Name: ");
    scanf(" %49[^\n]", r.name);
    printf("Enter Category (Medicine/Equipment/Bed/Supply): ");
    scanf(" %29[^\n]", r.category);
    printf("Enter Department: ");
    scanf(" %29[^\n]", r.department);
    printf("Enter Quantity Available: ");
    scanf("%d", &r.quantity);
    printf("Enter Minimum Threshold: ");
    scanf("%d", &r.threshold);
    printf("Enter Priority (1-High, 2-Medium, 3-Low): ");
    scanf("%d", &r.priority);

    resources[resCount++] = r;
    printf("Resource added successfully!\n");
}

void updateResource() {
    int id;
    printf("Enter Resource ID to update: ");
    scanf("%d", &id);

    int idx = searchByID(resources, resCount, id);
    if (idx == -1) {
        printf("Resource not found!\n");
        return;
    }

    Resource *r = &resources[idx];
    int choice;
    printf("1.Name 2.Category 3.Department 4.Quantity 5.Threshold 6.Priority\n");
    printf("Choose field to update: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: printf("New Name: ");        scanf(" %49[^\n]", r->name);        break;
        case 2: printf("New Category: ");    scanf(" %29[^\n]", r->category);    break;
        case 3: printf("New Department: ");  scanf(" %29[^\n]", r->department);  break;
        case 4: printf("New Quantity: ");    scanf("%d", &r->quantity);           break;
        case 5: printf("New Threshold: ");   scanf("%d", &r->threshold);          break;
        case 6: printf("New Priority: ");    scanf("%d", &r->priority);           break;
        default: printf("Invalid field choice.\n"); return;
    }

Appending to resource_management.c
```

Screenshot 4: *Cell 3 — swap(), bubbleSortByQuantity(), selectionSortByPriority(), sortByDepartment(), mergeDepartments()*

[3]

```
%%writefile -a resource_management.c

void swap(Resource *a, Resource *b) {
    Resource temp = *a;
    *a = *b;
    *b = temp;
}

void bubbleSortByQuantity(Resource *arr, int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if ((arr + j)->quantity > (arr + j + 1)->quantity) {
                swap(arr + j, arr + j + 1);
            }
        }
    }
}

void selectionSortByPriority(Resource *arr, int n) {
    for (int i = 0; i < n - 1; i++) {
        int minIdx = i;
        for (int j = i + 1; j < n; j++) {
            if ((arr + j)->priority < (arr + minIdx)->priority) {
                minIdx = j;
            }
        }
        if (minIdx != i) swap(arr + i, arr + minIdx);
    }
}

void sortByDepartment(Resource *arr, int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (strcmp((arr + j)->department, (arr + j + 1)->department) > 0) {
                swap(arr + j, arr + j + 1);
            }
        }
    }
}

void sortMenu() {
    int choice;
    printf("1.By Quantity  2.By Priority  3.By Department\nChoose sort criteria: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: bubbleSortByQuantity(resources, resCount);     break;
        case 2: selectionSortByPriority(resources, resCount);  break;
        case 3: sortByDepartment(resources, resCount);         break;
        default: printf("Invalid choice.\n"); return;
    }
    printf("Resources sorted successfully!\n");
    displayAll(resources, resCount);
}
```

```
Appending to resource_management.c
```

Screenshot 5: *Cell 4 — analyseResources(), displayCriticalResources(), generateReport(), file handling*

[4]

```c
%%writefile -a resource_management.c

void analyseResources() {
    int adequate = 0, low = 0, critical = 0;

    for (int i = 0; i < resCount; i++) {
        int diff = resources[i].quantity - resources[i].threshold;
        if (diff > 5)      adequate++;
        else if (diff >= 0) low++;
        else                critical++;
    }

    printf("\n--- Resource Availability Analysis ---\n");
    printf("Adequate Resources : %d\n", adequate);
    printf("Low Resources      : %d\n", low);
    printf("Critical Resources : %d\n", critical);
}

void displayCriticalResources() {
    int any = 0;
    printf("\n--- Critical / Low Stock Resources ---\n");
    for (int i = 0; i < resCount; i++) {
        int diff = resources[i].quantity - resources[i].threshold;
        if (diff < 5) {
            char *status = (diff < 0) ? "CRITICAL" : "LOW";
            printf("ID:%d  Name:%s  Dept:%s  Qty:%d  Threshold:%d  Status:%s\n",
                   resources[i].id, resources[i].name, resources[i].department,
                   resources[i].quantity, resources[i].threshold, status);
            any = 1;
        }
    }
    if (!any) printf("All resources are at adequate levels.\n");
}

double totalQuantityRecursive(Resource arr[], int n) {
    if (n <= 0) return 0;
    return arr[n - 1].quantity + totalQuantityRecursive(arr, n - 1);
}

void departmentWiseTotalRecursive(char depts[][30], int deptCount, int index) {
    if (index >= deptCount) return;

    int total = 0;
    for (int i = 0; i < resCount; i++) {
        if (strcmp(resources[i].department, depts[index]) == 0) {
            total += resources[i].quantity;
        }
    }
    printf("Department: %-15s Total Quantity: %d\n", depts[index], total);
    departmentWiseTotalRecursive(depts, deptCount, index + 1);
}

void generateReport() {
    FILE *fp = fopen(REPORTFILE, "w");
    if (fp == NULL) {
```

```
Appending to resource_management.c
```

Screenshot 6: *Cell 5 — compile with GCC and run: adding 5 resources across Emergency / ICU / Trauma departments*

```
!gcc -Wall -o resmgmt resource_management.c && ./resmgmt
```

[5]

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Enter Resource ID: Enter Resource Name: Enter Category (Medicine/Equipment/Bed/Supply):
Enter Department: Enter Quantity Available: Enter Minimum Threshold: Enter Priority (1-High, 2-Medium, 3-
Low): Resource added successfully!


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Enter Resource ID: Enter Resource Name: Enter Category (Medicine/Equipment/Bed/Supply):
Enter Department: Enter Quantity Available: Enter Minimum Threshold: Enter Priority (1-High, 2-Medium, 3-
Low): Resource added successfully!


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Enter Resource ID: Enter Resource Name: Enter Category (Medicine/Equipment/Bed/Supply):
Enter Department: Enter Quantity Available: Enter Minimum Threshold: Enter Priority (1-High, 2-Medium, 3-
Low): Resource added successfully!


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Enter Resource ID: Enter Resource Name: Enter Category (Medicine/Equipment/Bed/Supply):
Enter Department: Enter Quantity Available: Enter Minimum Threshold: Enter Priority (1-High, 2-Medium, 3-
Low): Resource added successfully!
```

Screenshot 7: *Cell 5 (continued) — duplicate-ID rejection, Display All, and Search by ID / Name / Category*

| output continued |

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Enter Resource ID: Enter Resource Name: Enter Category (Medicine/Equipment/Bed/Supply):
Enter Department: Enter Quantity Available: Enter Minimum Threshold: Enter Priority (1-High, 2-Medium, 3-
Low): Resource added successfully!


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Enter Resource ID: Error: a resource with this ID already exists!


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice:
ID   Name             Category   Department  Qty     Thresh    Prior
--------------------------------------------------------------------
101  Paracetamol      Medicine   Emergency   20      30        1
102  Oxygen Cylinder  Equipment  ICU         5       10        1
103  Hospital Bed     Bed        Emergency   15      5         2
104  Ventilator       Equipment  ICU         8       6         1
105  IV Fluid         Supply     Trauma      50      20        3
```

Screenshot 8: *Cell 5 (continued) — Sort by Quantity, Priority, and Department*

(output continued)

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: 1.By ID 2.By Name 3.By Category
Choose search type: Enter ID: Resource Found:

ID   Name              Category    Department  Qty     Thresh    Prior
-----------------------------------------------------------------
103  Hospital Bed      Bed         Emergency   15      5         2


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: 1.By ID 2.By Name 3.By Category
Choose search type: Enter Name: Resource Found:

ID   Name              Category    Department  Qty     Thresh    Prior
-----------------------------------------------------------------
104  Ventilator        Equipment   ICU         8       6         1
```

Screenshot 9: *Cell 5 (continued) — Merge Departments (duplicates removed) and invalid department merge*

(output continued)

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: 1.By ID 2.By Name 3.By Category
Choose search type: Enter Category: Found 2 resource(s):

ID   Name              Category    Department  Qty     Thresh   Prior
----------------------------------------------------------------
102  Oxygen Cylinder   Equipment   ICU         5       10       1

ID   Name              Category    Department  Qty     Thresh   Prior
----------------------------------------------------------------
104  Ventilator        Equipment   ICU         8       6        1


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: 1.By ID 2.By Name 3.By Category
Choose search type: Invalid choice.
```

Screenshot 10: *Cell 5 (continued) — Duplicate detection, Resource Analysis, and Critical/Low-Stock display*

[output continued]

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
==========================================================
Enter your choice: 1.By Quantity  2.By Priority  3.By Department
Choose sort criteria: Resources sorted successfully!

ID   Name             Category    Department  Qty    Thresh    Prior
--------------------------------------------------------------------
102  Oxygen Cylinder  Equipment   ICU         5      10        1
104  Ventilator       Equipment   ICU         8      6         1
103  Hospital Bed     Bed         Emergency   15     5         2
101  Paracetamol      Medicine    Emergency   20     30        1
105  IV Fluid         Supply      Trauma      50     20        3

===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
==========================================================
Enter your choice: 1.By Quantity  2.By Priority  3.By Department
Choose sort criteria: Resources sorted successfully!

ID   Name             Category    Department  Qty    Thresh    Prior
--------------------------------------------------------------------
102  Oxygen Cylinder  Equipment   ICU         5      10        1
104  Ventilator       Equipment   ICU         8      6         1
101  Paracetamol      Medicine    Emergency   20     30        1
103  Hospital Bed     Bed         Emergency   15     5         2
105  IV Fluid         Supply      Trauma      50     20        3
```

Screenshot 11: *Cell 5 (continued) — Generate Consolidated Report, Save/Retrieve from file*

[output continued]

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: 1.By Quantity  2.By Priority  3.By Department
Choose sort criteria: Resources sorted successfully!

ID   Name              Category    Department  Qty     Thresh    Prior
-----------------------------------------------------------------
101  Paracetamol       Medicine    Emergency   20      30        1
103  Hospital Bed      Bed         Emergency   15      5         2
102  Oxygen Cylinder   Equipment   ICU         5       10        1
104  Ventilator        Equipment   ICU         8       6         1
105  IV Fluid          Supply      Trauma      50      20        3


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: Enter first department name: Enter second department name:
Merged Resource List (Emergency + ICU), duplicates removed:

ID   Name              Category    Department  Qty     Thresh    Prior
-----------------------------------------------------------------
101  Paracetamol       Medicine    Emergency   20      30        1
103  Hospital Bed      Bed         Emergency   15      5         2
102  Oxygen Cylinder   Equipment   ICU         5       10        1
104  Ventilator        Equipment   ICU         8       6         1


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: Enter first department name: Enter second department name: No resources found for the
given departments.
```

Screenshot 12: *Cell 5 (continued) — invalid menu inputs and program Exit*

[output continued]

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: No duplicate resources found.


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice:
--- Resource Availability Analysis ---
Adequate Resources : 2
Low Resources      : 1
Critical Resources : 2


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice:
--- Critical / Low Stock Resources ---
ID:101  Name:Paracetamol  Dept:Emergency  Qty:20  Threshold:30  Status:CRITICAL
ID:102  Name:Oxygen Cylinder  Dept:ICU  Qty:5  Threshold:10  Status:CRITICAL
ID:104  Name:Ventilator  Dept:ICU  Qty:8  Threshold:6  Status:LOW
```

Screenshot 13: *Cell 5 (continued) — output part 8*

[output continued]

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: 
========= CONSOLIDATED RESOURCE REPORT =========
Total Resources Available (all departments): 98

Department-wise Availability:
Department: Emergency       Total Quantity: 35
Department: ICU             Total Quantity: 13
Department: Trauma          Total Quantity: 50

Most Critical Resources (immediate replenishment needed):
ID:101 Name:Paracetamol Dept:Emergency Qty:20 (Needed:30)
ID:102 Name:Oxygen Cylinder Dept:ICU Qty:5 (Needed:10)

Report generated and saved to report.txt


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: Records saved to file successfully!


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
========================================================
Enter your choice: Records loaded from file successfully! (5 records)
```

Screenshot 14: *Cell 5 (continued) — output part 9*

[output continued]

```
===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Invalid choice! Please try again.


===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Invalid input! Please enter a number.

===== SMART EMERGENCY RESOURCE MANAGEMENT SYSTEM =====
 1. Add New Resource
 2. Update Resource Details
 3. Display All Resources
 4. Search Resource (ID / Name / Category)
 5. Sort Resources (Quantity / Priority / Department)
 6. Merge Resources from Two Departments
 7. Identify Duplicate Resources
 8. Analyse Resource Availability
 9. Display Critical / Low-Stock Resources
10. Generate Consolidated Report
11. Save Records to File
12. Retrieve Records from File
13. Exit
=========================================================
Enter your choice: Saving data and exiting...
Records saved to file successfully!
```